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JEE MAIN · QUESTION PAPER

JEE Main 2026 (Online) 21st January Morning Shift

JEE Main Question Paper with Answer Key

21 Jan 2026	75	English
HELD ON	QUESTIONS	LANGUAGE

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

Open solution

Consider the following reactions.

$$\underset{\text{(Hot solution)}}{\mathrm{PbCl_2}} + \mathrm{K_2CrO_4} \rightarrow A + 2\mathrm{KCl}$$

$$\mathrm{A} + \mathrm{NaOH} \rightleftharpoons \mathrm{B} + \mathrm{Na_2CrO_4}$$

$$\mathrm{PbSO_4} + 4\mathrm{CH_3COONH_4} \rightarrow (\mathrm{NH_4})_2\mathrm{SO_4} + \mathrm{X}$$

In the above reactions, A, B and X are respectively.

(A) $\mathrm{Na_2}\left[\mathrm{Pb(OH)_2}\right], \mathrm{PbCrO_4}$ and $\mathrm{[Pb(NH_3)_4]SO_4}$

(B) $\mathrm{Na_2}\left[\mathrm{Pb(OH)_2}\right], \mathrm{PbCrO_4}$ and $(\mathrm{NH_4})_2\left[\mathrm{Pb(CH_3COO)_4}\right]$

(C) $\mathrm{PbCrO_4}, \mathrm{Na_2}\left[\mathrm{Pb(OH)_4}\right]$ and $(\mathrm{NH_4})_2\left[\mathrm{Pb(CH_3COO)_4}\right]$

(D) $\mathrm{PbCrO_4}, \mathrm{Na_2}\left[\mathrm{Pb(OH)_4}\right]$ and $\mathrm{[Pb(NH_3)_4]SO_4}$

Q2. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Among $\mathrm{[Cu(NH_3)_4]^{2+}}, \mathrm{[Ni(en)_3]^{2+}}, \mathrm{[Ni(NH_3)_6]^{2+}}$ and $\mathrm{[Mn(H_2O)_6]^{2+}}, \mathrm{[Mn(H_2O)_6]^{2+}}$ has the maximum number of unpaired electrons.

Statement II : The number of pairs among $\left\{\mathrm{[NiCl_4]^{2-}}, \mathrm{[Ni(CO)_4]}\right\}, \left\{\mathrm{[NiCl_4]^{2-}}, \mathrm{[Ni(CN)_4]^{2-}}\right\}$ and $\left\{\mathrm{[Ni(CO)_4]}, \mathrm{[Ni(CN)_4]^{2-}}\right\}$ that contain only diamagnetic species is two.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q3. MCQ (Single Correct Answer)

Open solution

An organic compound (P) on treatment with aqueous ammonia under hot condition forms compound (Q) which on heating with $\mathrm{Br_2}$ and KOH forms compound (R) having molecular formula $\mathrm{C_6H_7N}$. Names of P, Q and R respectively are.

(A) Toluic acid, methylbenzamide, 2-methylaniline

(B) Benzoic acid, benzamide, aniline

(C) Benzoic acid, 4-methylbenzamide, 4-methylaniline

(D) Phenylethanoic acid, phenylethanamide, benzamine

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Q4. MCQ (Single Correct Answer)

Open solution

From the following, the least stable structure is :

(A)



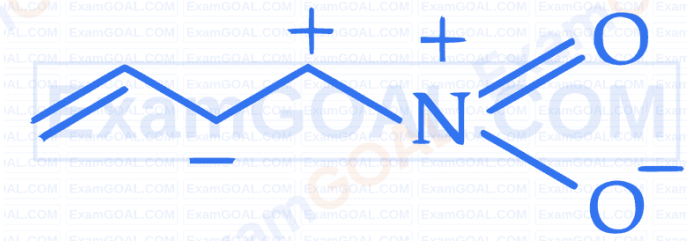
(B)



(C)



(D)



Q5. MCQ (Single Correct Answer)

Open solution

Identify the correct statements.

A. Arginine and Tryptophan are essential amino acids.

B. Histidine does not contain heterocyclic ring in its structure.

C. Proline is a six membered cyclic ring amino acid.

D. Glycine does not have chiral centre.

E. Cysteine has characteristic feature of side chain as $\mathrm{MeS} - \mathrm{CH_2} - \mathrm{CH_2} -$.

Choose the correct answer from the options given below :

(A) C and D Only (B) A and D Only (C) C and E Only (D) B and E Only

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Q6. MCQ (Single Correct Answer)

Open solution

A hydrocarbon ' P ' (C_4H_8) on reaction with HCl gives an optically active compound ' Q ' (C_4H_9Cl) which on reaction with one mole of ammonia gives compound ' R ' $(C_4H_{11}N)$. ' R ' on diazotization followed by hydrolysis gives ' S '. Identify P, Q, R and S .

(A) $P=CH_3-CH=CH-CH_3,\ Q=CH_3-CH_2-CH_2-CH_2-Cl,\ R=CH_3-CH_2-CH_2-\underset{NH_2}{CH_2},\ S=CH_3-CH_2-\underset{OH}{\underset{|}{CH_2}}$

(B) $P=CH_3-CH=CH-CH_3,\ Q=CH_3-CH_2-\underset{Cl}{\underset{|}{CH}}-CH_3,\ R=CH_3-CH_2-\underset{NH_2}{\underset{|}{CH}}-CH_3,\ S=CH_3-CH_2-\underset{OH}{\underset{|}{CH}}-CH_3$

(C) $P=CH_3$  , $Q=Cl-CH_2$  , $R=H_2N-CH_2$  , $S=$ 

(D) $P=CH_3-CH_2-CH=CH_2,\ Q=CH_3-CH_2-CH_2-\underset{Cl}{\underset{|}{CH_2}},\ R=CH_3-CH_2-CH_2-\underset{CH_3}{\underset{|}{NH}},\ S=CH_3-CH_2-\underset{CH_3}{\underset{|}{\overset{OH}{\overset{|}{CH}}}}$

Q7. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The number of pairs among $[SiO_2, CO_2]$, $[SnO, SnO_2]$, $[PbO, PbO_2]$ and $[GeO, GeO_2]$, which contain oxides that are both amphoteric is 2 .

Statement II : BF_3 is an electron deficient molecule, can act as a Lewis acid, forms adduct with NH_3 and has a trigonal planar geometry.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

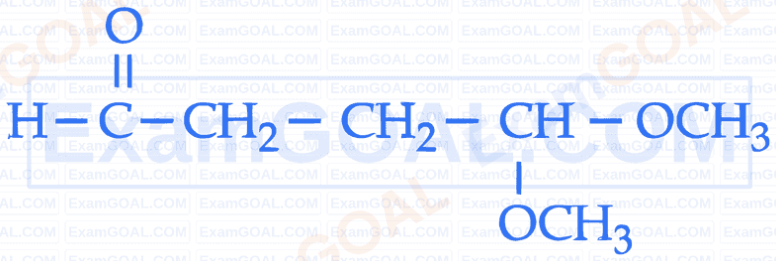
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Q8. MCQ (Single Correct Answer)

Open solution

An organic compound " P " of molecular formula $C_6H_{12}O_3$ gives positive Iodoform test but negative Tollen's test. When " P " is treated with dilute acid, it produces " Q ". " Q " gives positive Tollen's test and also iodoform test. The structure of " P " is :

(A)



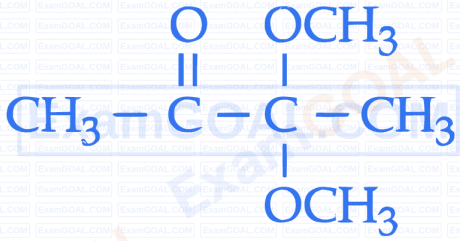
(B)



(C)



(D)



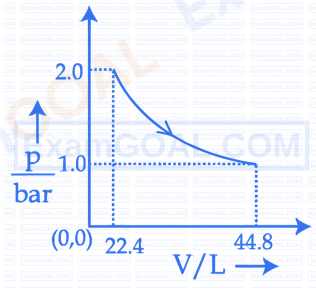
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Q9. MCQ (Single Correct Answer)

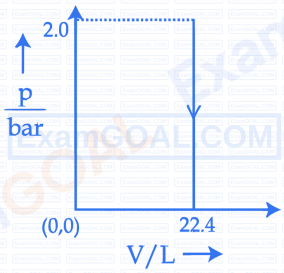
Which of the following graphs between pressure ' p ' versus volume ' V ' represents the maximum work done?

Open solution

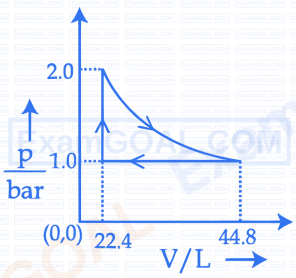
(A)



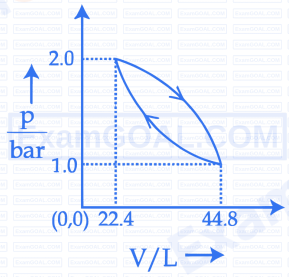
(B)



(C)



(D)



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Q10. MCQ (Single Correct Answer)

Open solution

For the reaction, $N_2O_4 \rightleftharpoons 2NO_2$, graph is plotted as shown below. Identify correct statements.

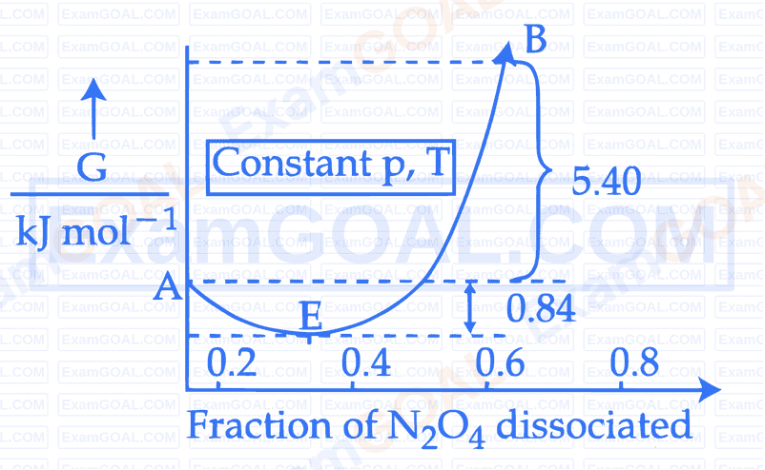
A. Standard free energy change for the reaction is $-5.40\,kJ\,mol^{-1}$.

B. As $\Delta G^{\ominus}$ in graph is positive, N_2O_4 will not dissociate into NO_2 at all.

C. Reverse reaction will go to completion.

D. When 1 mole of N_2O_4 changes into equilibrium mixture, value of $\Delta G^{\ominus} = -0.84\,kJ\,mol^{-1}$

E. When 2 mole of NO_2 changes into equilibrium mixture, $\Delta G^{\ominus}$ for equilibrium mixture is $-6.24\,kJ\,mol^{-1}$.



Choose the correct answer from the options given below :

(A) C and E only (B) D and E only (C) A and D only (D) B and C only

Q11. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I :

The number of species among $SF_4, NH_4^+, [NiCl_4]^{2-}, XeF_4, [PtCl_4]^{2-}, SeF_4$ and $[Ni(CN)_4]^{2-}$, that have tetrahedral geometry is 3 .

Statement II :

In the set [$NO_2, BeH_2, BF_3, AlCl_3$], all the molecules have incomplete octet around central atom.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

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Q12. MCQ (Single Correct Answer)

Open solution

Identify **correct** statements from the following :

A. Propanal and propanone are functional isomers.

B. Ethoxyethane and methoxypropane are metamers.

C. But-2-ene shows optical isomerism.

D. But-1-ene and but-2-ene are functional isomers.

E. Pentane and 2, 2-dimethyl propane are chain isomers.

Choose the **correct** answer from the options given below :

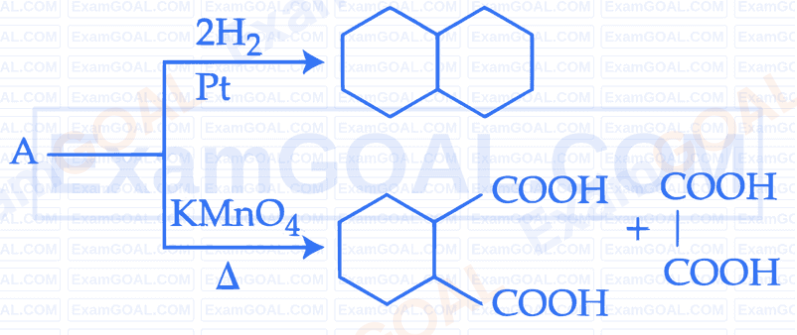
(A) A, B and E only **(B)** A, B and C only **(C)** C, D and E only **(D)** B, C and D only

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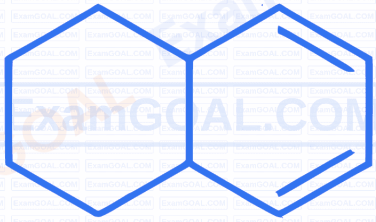
Q13. MCQ (Single Correct Answer)

Open solution

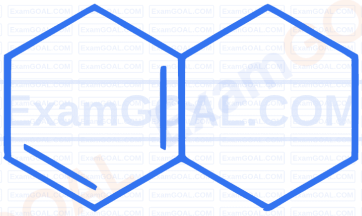
Identify A in the following reaction.



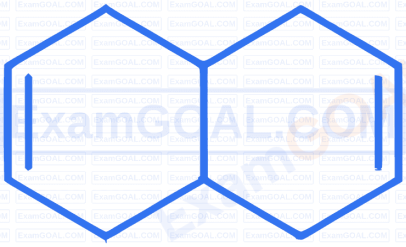
(A)



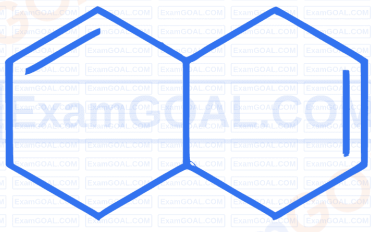
(B)



(C)



(D)



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Q14. MCQ (Single Correct Answer)

14.0 g of calcium metal is allowed to react with excess HCl at 1.0 atm pressure and 273 K . Which of the following statements is incorrect?

[Given : Molar mass in gmol^{-1} of $\mathrm{Ca}-40, \mathrm{Cl}-35.5, \mathrm{H}-1$]

(A) The limiting reagent is calcium metal.

(B) 33.3 g of CaCl_2 is produced.

(C) 7.84 L of H_2 gas is evolved.

(D) 0.35 mol of H_2 gas is evolved.

Open solution

Q15. MCQ (Single Correct Answer)

Which of the following represents the correct trend for the mentioned property?

A. F > P > S > B - First Ionization Energy

B. Cl > F > S > P - Electron Affinity

C. K > Al > Mg > B - Metallic character

D. $K_2O > Na_2O > MgO > Al_2O_3$ - Basic character

Choose the **correct** answer from the options given below :

(A) B and C only

(B) A, B and D only

(C) A and B only

(D) A, B, C and D

Open solution

Q16. MCQ (Single Correct Answer)

80 mL of a hydrocarbon on mixing with 264 mL of oxygen in a closed U-tube undergoes complete combustion. The residual gases after cooling to 273 K occupy 224 mL . When the system is treated with KOH solution, the volume decreases to 64 mL . The formula of the hydrocarbon is :

(A) $\mathrm{C}_2\mathrm{H}_2$

(B) $\mathrm{C}_2\mathrm{H}_4$

(C) $\mathrm{C}_4\mathrm{H}_{10}$

(D) $\mathrm{C}_2\mathrm{H}_6$

Open solution

Q17. MCQ (Single Correct Answer)

$\mathrm{MnO_4}^{2-}$, in acidic medium, disproportionates to :

(A) MnO_4^- and MnO

(B) MnO_4^- and MnO_2

(C) $\mathrm{Mn}_2\mathrm{O}_7$ and MnO_2

(D) $\mathrm{Mn}_2\mathrm{O}_7$ and MnO

Open solution

Q18. MCQ (Single Correct Answer)

In Carius method, 0.75 g of an organic compound gave 1.2 g of barium sulphate, find percentage of sulphur (molar mass $32\ \mathrm{g\ mol}^{-1}$). Molar mass of barium sulphate is $233\ \mathrm{g\ mol}^{-1}$.

(A) 4.55%

(B) 16.48%

(C) 10.30%

(D) 21.97%

Open solution

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Q19. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : When an electric discharge is passed through gaseous hydrogen, the hydrogen molecules dissociate and the energetically excited hydrogen atoms produce electromagnetic radiation of discrete frequencies.

Statement II : The frequency of second line of Balmer series obtained from He^+ is equal to that of first line of Lyman series obtained from hydrogen atom.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q20. MCQ (Single Correct Answer)

Open solution

Elements P and Q form two types of non-volatile, non-ionizable compounds PQ and PQ_2. When 1 g of PQ is dissolved in 50 g of solvent ' A', ΔT_b was 1.176 K while when 1 g of PQ_2 is dissolved in 50 g of solvent ' A', ΔT_b was 0.689 K . (K_b of ' A' $= 5\ K\ kg\ mol^{-1}$). The molar masses of elements P and Q (in $gmol^{-1}$) respectively, are :

(A) 25, 60 (B) 60, 25 (C) 65, 145 (D) 70, 110

Q21. Numerical

Open solution

Pre-exponential factors of two different reactions of same order are identical. Let activation energy of first reaction exceeds the activation energy of second reaction by $20\ kJ\ mol^{-1}$. If k_1 and k_2 are the rate constants of first and second reaction respectively at 300 K , then $\ln \frac{k_2}{k_1}$ will be ____ . (nearest integer) $\left[R = 8.3\ J\ K^{-1}\ mol^{-1}\right]$

Q22. Numerical

Open solution

Use the following data :

Substance	$\frac{\Delta_f H^\ominus (500\ K)}{kJ\ mol^{-1}}$	$\frac{S^\ominus (500\ K)}{JK^{-1}\ mol^{-1}}$
$AB(g)$	32	222
$A_2(g)$	6	146
$B_2(g)$	x	280

One mole each of $A_2(g)$ and $B_2(g)$ are taken in a 1 L closed flask and allowed to establish the equilibrium at 500 K .

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

The value of x $(inkJmol^{-1})$ is ____ . (Nearest integer)

(Given : $\log K = 2.2 \quad R = 8.3\ J\ K^{-1}\ mol^{-1}$)

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Q23. Numerical

Consider the following reactions :

$$NaCl + K_2Cr_2O_7 + H_2SO_4 \rightarrow A + KHSO_4 + NaHSO_4 + H_2O$$
$$A + NaOH \rightarrow B + NaCl + H_2O$$
$$B + H_2SO_4 + H_2O_2 \rightarrow C + Na_2SO_4 + H_2O$$

In the product ' C', ' X ' is the number of O_2^{2-} units, ' Y ' is the total number of oxygen atoms present and ' Z ' is the oxidation state of Cr. The value of $X + Y + Z$ is ____ .

Open solution

Q24. Numerical

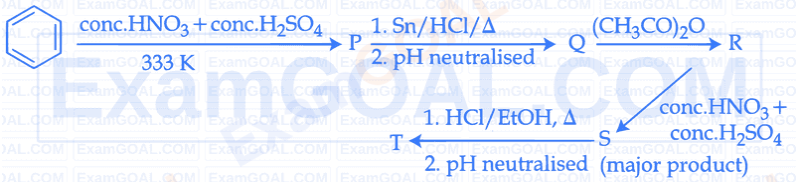
The pH and conductance of a weak acid (HX) was found to be 5 and 4×10^{-5} S, respectively. The conductance was measured under standard condition using a cell where the electrode plates having a surface area of $1\ cm^2$ were at a distance of 15 cm apart. The value of the limiting molar conductivity is ____ $Sm^2\ mol^{-1}$. (nearest integer)

(Given : degree of dissociation of the weak acid $(\alpha) \ll 1$)

Open solution

Q25. Numerical

Consider the following reaction sequence



The percentage of nitrogen in product ' T ' formed is ____ %. (Nearest integer)

(Given molar mass in $gmol^{-1}$ $H : 1, C : 12, N : 14, O : 16$)

Open solution

Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let the mean and variance of 7 observations $2, 4, 10, x, 12, 14, y, x > y$, be 8 and 16 respectively. Two numbers are chosen from $\{1, 2, 3, x-4, y, 5\}$ one after another without replacement, then the probability, that the smaller number among the two chosen numbers is less than 4 , is :

(A) $\frac{4}{5}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{1}{3}$

Open solution

Q27. MCQ (Single Correct Answer)

If $x^2 + x + 1 = 0$, then the value of $\left(x + \frac{1}{x}\right)^4 + \left(x^2 + \frac{1}{x^2}\right)^4 + \left(x^3 + \frac{1}{x^3}\right)^4 + \ldots + \left(x^{25} + \frac{1}{x^{25}}\right)^4$ is:

(A) 162 (B) 145 (C) 128 (D) 175

Open solution

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Q28. MCQ (Single Correct Answer)

Let a point A lie between the parallel lines L_1 and L_2 such that its distances from L_1 and L_2 are 6 and 3 units, respectively. Then the area (in sq. units) of the equilateral triangle ABC , where the points B and C lie on the lines L_1 and L_2, respectively, is :

(A) $21\sqrt{3}$ (B) $12\sqrt{2}$ (C) $15\sqrt{6}$ (D) 27

Open solution

Q29. MCQ (Single Correct Answer)

The value of $\operatorname{cosec} 10^\circ - \sqrt{3}\sec 10^\circ$ is equal to :

(A) 2 (B) 6 (C) 8 (D) 4

Open solution

Q30. MCQ (Single Correct Answer)

Let $\vec{a} = -\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 8\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} = \vec{b}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, then $|\vec{a} + \vec{c}|^2$ is equal to :

(A) 30 (B) 33 (C) 27 (D) 35

Open solution

Q31. MCQ (Single Correct Answer)

The value of $\int_{-\pi/6}^{\pi/6} \left(\frac{\pi + 4x^{11}}{1 - \sin(|x| + \pi/6)}\right) dx$ is equal to:

(A) 8π (B) 4π (C) 2π (D) 6π

Open solution

Q32. MCQ (Single Correct Answer)

Let $y = y(x)$ be the solution curve of the differential equation $(1 + x^2)dy + (y - \tan^{-1} x)dx = 0, y(0) = 1$. Then the value of $y(1)$ is :

(A) $\frac{4}{e^{\pi/4}} - \frac{\pi}{2} - 1$ (B) $\frac{2}{e^{\pi/4}} + \frac{\pi}{4} - 1$

(C) $\frac{4}{e^{\pi/4}} + \frac{\pi}{2} - 1$ (D) $\frac{2}{e^{\pi/4}} - \frac{\pi}{4} - 1$

Open solution

Q33. MCQ (Single Correct Answer)

The number of strictly increasing functions f from the set $\{1, 2, 3, 4, 5, 6\}$ to the set $\{1, 2, 3, \ldots, 9\}$ such that $f(i) \neq i$ for $1 \leq i \leq 6$, is equal to :

(A) 21 (B) 28 (C) 27 (D) 22

Open solution

Q34. MCQ (Single Correct Answer)

Let (α, β, γ) be the co-ordinates of the foot of the perpendicular drawn from the point $(5, 4, 2)$ on the line $\vec{r} = (-\hat{i} + 3\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$.

Then the length of the projection of the vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ on the vector $6\hat{i} + 2\hat{j} + 3\hat{k}$ is :

(A) $\frac{18}{7}$ (B) $\frac{15}{7}$ (C) 4 (D) 3

Open solution

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Q35. MCQ (Single Correct Answer)

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms such that $a_2 \cdot a_3 \cdot a_4 = 64$ and $a_1 + a_3 + a_5 = \frac{813}{7}$. Then $a_3 + a_5 + a_7$ is equal to :

(A) 3256 (B) 3252 (C) 3248 (D) 3244

Open solution

Q36. MCQ (Single Correct Answer)

Let PQ and MN be two straight lines touching the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ at the points A and B respectively. Let O be the centre of the circle and $\angle AOB = \pi/3$. Then the locus of the point of intersection of the lines PQ and MN is :

(A) $x^2 + y^2 - 18x - 12y - 25 = 0$ (B) $x^2 + y^2 - 12x - 18y - 25 = 0$

(C) $3\left(x^2 + y^2\right) - 12x - 18y - 25 = 0$ (D) $3\left(x^2 + y^2\right) - 18x - 12y + 25 = 0$

Open solution

Q37. MCQ (Single Correct Answer)

The area of the region, inside the ellipse $x^2 + 4y^2 = 4$ and outside the region bounded by the curves $y = |x| - 1$ and $y = 1 - |x|$, is :

(A) $2\pi - 1$ (B) $3(\pi - 1)$ (C) $2(\pi - 1)$ (D) $2\pi - \frac{1}{2}$

Open solution

Q38. MCQ (Single Correct Answer)

The sum of all the roots of the equation $(x-1)^2 - 5|x-1| + 6 = 0$, is :

(A) 3 (B) 1 (C) 4 (D) 5

Open solution

Q39. MCQ (Single Correct Answer)

If the coefficient of x in the expansion of $\left(ax^2 + bx + c\right)(1 - 2x)^{26}$ is -56 and the coefficients of x^2 and x^3 are both zero, then $\mathrm{a} + \mathrm{b} + \mathrm{c}$ is equal to :

(A) 1483 (B) 1300 (C) 1500 (D) 1403

Open solution

Q40. MCQ (Single Correct Answer)

The number of relations, defined on the set $\{\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}\}$, which are both reflexive and symmetric, is equal to:

(A) 16 (B) 64 (C) 256 (D) 1024

Open solution

Q41. MCQ (Single Correct Answer)

Let the foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If the eccentricity of the hyperbola is 5 , then the length of its latus rectum is :

(A) $\frac{96}{\sqrt{5}}$ (B) $24\sqrt{5}$ (C) 12 (D) 16

Open solution

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Q42. MCQ (Single Correct Answer)

If the domain of the function $f(x)=\cos^{-1}\left(\frac{2x-5}{11-3x}\right)+\sin^{-1}\left(2x^2-3x+1\right)$ is the interval $[\alpha,\beta]$, then $\alpha+2\beta$ is equal to :

(A) 5 (B) 2 (C) 3 (D) 1

Open solution

Q43. MCQ (Single Correct Answer)

Let O be the vertex of the parabola $x^2=4y$ and Q be any point on it. Let the locus of the point P , which divides the line segment OQ internally in the ratio $2:3$ be the conic C . Then the equation of the chord of C, which is bisected at the point $(1,2)$, is :

(A) $5x-4y+3=0$ (B) $x-2y+3=0$ (C) $4x-5y+6=0$ (D) $5x-y-3=0$

Open solution

Q44. MCQ (Single Correct Answer)

Let $\overrightarrow{c}$ and $\overrightarrow{d}$ be vectors such that $|\overrightarrow{c}+\overrightarrow{d}|=\sqrt{29}$ and $\overrightarrow{c}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\overrightarrow{d}$. If $\lambda_1,\lambda_2\,(\lambda_1>\lambda_2)$ are the possible values of $(\vec{c}+\vec{d})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})$, then the equation $\mathrm{K}^2x^2+\left(\mathrm{K}^2-5\,\mathrm{K}+\lambda_1\right)xy+\left(3\,\mathrm{K}+\frac{\lambda_2}{2}\right)y^2-8x+12y+\lambda_2=0$ represents a circle, for K equal to :

(A) 4 (B) -1 (C) 2 (D) 1

Open solution

Q45. MCQ (Single Correct Answer)

Let $f:\mathbf{R}\to(0,\infty)$ be a twice differentiable function such that $f(3)=18, f'(3)=0$ and $f''(3)=4$.

Then $\lim\limits_{x\to1}\left(\log_e\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}\right)$ is equal to :

(A) 9 (B) 18 (C) 1 (D) 2

Open solution

Q46. Numerical

Let $f:\mathbf{R}\to\mathbf{R}$ be a twice differentiable function such that the quadratic equation $f(x)\mathrm{m}^2-2f'(x)\mathrm{m}+f''(x)=0$ in m , has two equal roots for every $x\in\mathbf{R}$. If $f(0)=1, f'(0)=2$, and (α,β) is the largest interval in which the function $f\left(\log_e x-x\right)$ is increasing, then $\alpha+\beta$ is equal to ____ .

Open solution

Q47. Numerical

Let $S=\{(m,n):m,n\in\{1,2,3,\ldots\ldots,50\}\}$. If the number of elements (m,n) in S such that 6^m+9^n is a multiple of 5 is p and the number of elements (m,n) in S such that $m+n$ is a square of a prime number is q , then $\mathrm{p}+\mathrm{q}$ is equal to ____ .

Open solution

Q48. Numerical

Let $a_1=1$ and for $n\geqslant1, a_{n+1}=\frac{1}{2}a_n+\frac{n^2-2n-1}{n^2(n+1)^2}$. Then $\left|\sum_{n=1}^{\infty}\left(a_n-\frac{2}{n^2}\right)\right|$ is equal to ____ .

Open solution

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Q49. Numerical

$6\int_0^{\pi}|(\sin 3x+\sin 2x+\sin x)|dx$ is equal to ____ .

Open solution

Q50. Numerical

For some $\alpha,\beta\in\mathbf{R}$, let $A=\begin{bmatrix}\alpha & 2\\ 1 & 2\end{bmatrix}$ and $B=\begin{bmatrix}1 & 1\\ 1 & \beta\end{bmatrix}$ be such that $A^2-4A+2I=B^2-3B+I=O$. Then $\left(\det\left(\text{adj}\left(A^3-B^3\right)\right)\right)^2$ is equal to ____ .

Open solution

Physics 25 questions

Q51. MCQ (Single Correct Answer)

A conducting circular loop of area $1.0\,\text{m}^2$ is placed perpendicular to a magnetic field which varies as $B=\sin(100t)$ Tesla. If the resistance of the loop is 100Ω, then the average thermal energy dissipated in the loop in one period is ____ J.

Open solution

(A) π^2 (B) $\frac{\pi}{2}$ (C) π (D) 2π

Q52. MCQ (Single Correct Answer)

A 4 kg mass moves under the influence of a force $\vec{F}=\left(4t^3\hat{i}-3t\hat{j}\right)\text{N}$ where t is the time in second. If mass starts from origin at $t=0$, the velocity and position after $t=2\,\text{s}$ will be:

Open solution

(A) $\vec{v}=3\hat{i}+\frac{3}{2}\hat{j}\quad\vec{r}=\frac{6}{5}\hat{i}+\hat{j}$

(B) $\vec{v}=4\hat{i}-\frac{3}{2}\hat{j}\quad\vec{r}=\frac{6}{5}\hat{i}-\hat{j}$

(C) $\vec{v}=4\hat{i}-\frac{3}{2}\hat{j}\quad\vec{r}=\frac{8}{5}\hat{i}-\hat{j}$

(D) $\vec{v}=4\hat{i}+\frac{5}{2}\hat{j}\quad\vec{r}=\frac{8}{5}\hat{i}+2\hat{j}$

Q53. MCQ (Single Correct Answer)

In an experiment the values of two spring constants were measured as $k_1=(10\pm0.2)\text{N/m}$ and $k_2=(20\pm0.3)\text{N/m}$. If these springs are connected in parallel, then the percentage error in equivalent spring constant is :

Open solution

(A) 1.33% (B) 2.67% (C) 1.67% (D) 2.33%

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Q54. MCQ (Single Correct Answer)

Open solution

Consider a modified Bernoulli equation.

$\left(\mathrm{P}+\frac{A}{Bt^2}\right)+\rho g(h+Bt)+\frac{1}{2}\rho V^2=$ constant

If t has the dimension of time then the dimensions of A and B are ____ , ____ respectively.

(A) $\left[\mathrm{ML^0\,T^{-1}}\right]$ and $\left[\mathrm{M^0LT}\right]$

(B) $\left[\mathrm{ML^0\,T^{-2}}\right]$ and $\left[\mathrm{M^0LT^{-1}}\right]$

(C) $\left[\mathrm{ML^0\,T^{-2}}\right]$ and $\left[\mathrm{M^0LT^{-2}}\right]$

(D) $\left[\mathrm{ML^0\,T^{-1}}\right]$ and $\left[\mathrm{M^0LT^{-1}}\right]$

Q55. MCQ (Single Correct Answer)

Open solution

An aluminium and steel rods having same lengths and cross-sections are joined to make total length of 120 cm at $30^\circ\mathrm{C}$. The coefficient of linear expansion of aluminium and steel are $24\times10^{-6}/^\circ\mathrm{C}$ and $1.2\times10^{-5}/^\circ\mathrm{C}$, respectively. The length of this composite rod when its temperature is raised to $100^\circ\mathrm{C}$, is ____ cm.

(A) 120.20 (B) 120.06 (C) 120.15 (D) 120.03

Q56. MCQ (Single Correct Answer)

Open solution

Two strings (A, B) having linear densities $\mu_A=2\times10^{-4}\,\mathrm{kg/m}$ and, $\mu_B=4\times10^{-4}\,\mathrm{kg/m}$ and lengths $L_A=2.5\,\mathrm{m}$ and $L_B=1.5\,\mathrm{m}$ respectively are joined. Free ends of A and B are tied to two rigid supports C and D, respectively creating a tension of 500 N in the wire. Two identical pulses, sent from C and D ends, take time t_1 and t_2, respectively, to reach the joint. The ratio t_1/t_2 is:

(A) 1.90 (B) 1.18 (C) 1.08 (D) 1.67

Q57. MCQ (Single Correct Answer)

Open solution

The electric field in a plane electromagnetic wave is given by :

$E_y=69\sin\left[0.6\times10^3x-1.8\times10^{11}t\right]\mathrm{V/m}.$

The expression for magnetic field associated with this electromagnetic wave is ____ T.

(A) $B_z=2.3\times10^{-7}\sin\left[0.6\times10^3x-1.8\times10^{11}t\right]$

(B) $B_z=2.3\times10^{-7}\sin\left[0.6\times10^3x+1.8\times10^{11}t\right]$

(C) $B_y=2.3\times10^{-7}\sin\left[0.6\times10^3x-1.8\times10^{11}t\right]$

(D) $B_y=69\sin\left[0.6\times10^3x+1.8\times10^{11}t\right]$

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Q58. MCQ (Single Correct Answer)

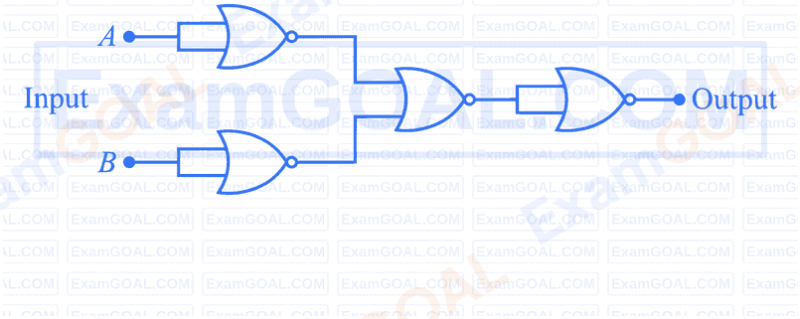
A parallel plate capacitor has capacitance C, when there is vacuum within the parallel plates. A sheet having thickness $\left(\frac{1}{3}\right)^{\text{rd}}$ of the separation between the plates and relative permittivity K is introduced between the plates. The new capacitance of the system is :

Open solution

(A) $\frac{3CK^2}{(2K+1)^2}$ (B) $\frac{4KC}{3K-1}$ (C) $\frac{CK}{2+K}$ (D) $\frac{3KC}{2K+1}$

Q59. MCQ (Single Correct Answer)

The given circuit works as :



Open solution

(A) NOR gate (B) OR gate (C) AND gate (D) NAND gate

Q60. MCQ (Single Correct Answer)

A point charge of 10^{-8}C is placed at origin. The work done in moving a point charge $2\mu\text{C}$ from point $A(4,4,2)\text{m}$ to $B(2,2,1)\text{m}$ is ____ J. $\left(\frac{1}{4\pi\epsilon_o} = 9\times10^9 \text{ in SI units}\right)$

Open solution

(A) 30×10^{-6} (B) 0 (C) 15×10^{-6} (D) 45×10^{-6}

Q61. MCQ (Single Correct Answer)

A light wave described by $E = 60\left[\sin\left(3\times10^{15}\right)t + \sin\left(12\times10^{15}\right)t\right]$ (in SI units) falls on a metal surface of work function 2.8 eV . The maximum kinetic energy of ejected photoelectron is (approximately)

____ eV. $\left(h = 6.6\times10^{-34}\text{ J.s. and } e = 1.6\times10^{-19}\text{C}\right)$

Open solution

(A) 3.8 (B) 7.8 (C) 6.0 (D) 5.1

Q62. MCQ (Single Correct Answer)

Initially a satellite of 100 kg is in a circular orbit of radius 1.5R_E. This satellite can be moved to a circular orbit of radius $3R_E$ by supplying $\alpha\times10^6$ J of energy. The value of α is ____.

(Take Radius of Earth $R_E = 6\times10^6$ m and $\text{g} = 10\,\text{m/s}^2$)

Open solution

(A) 500 (B) 1000 (C) 100 (D) 150

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Q63. MCQ (Single Correct Answer)

A gas based geyser heats water flowing at the rate of 5.0 litres per minute from $27^{\circ}\mathrm{C}$ to $87^{\circ}\mathrm{C}$. The rate of consumption of the gas is ____ $\mathrm{g/s}$.

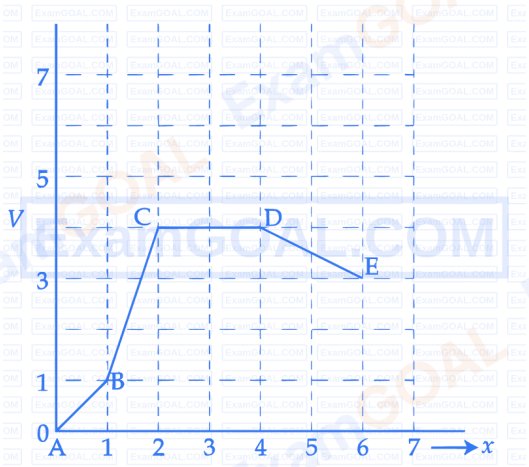
(Take heat of combustion of gas $= 5.0 \times 10^4\ \mathrm{J/g}$, specific heat capacity of water $= 4200\ \mathrm{J/kg \cdot {}^{\circ}C}$)

Open solution

(A) 4.2 (B) 2.1 (C) 0.21 (D) 0.42

Q64. MCQ (Single Correct Answer)

Potential energy (V) versus distance (x) is given by the graph. Rank various regions as per the magnitudes of the force (F) acting on a particle from high to low.



Open solution

(A) $F_{BC} > F_{AB} > F_{DE} > F_{CD}$ (B) $F_{BC} > F_{CD} > F_{DE} > F_{AB}$

(C) $F_{CD} > F_{AB} > F_{BC} > F_{DE}$ (D) $F_{CD} > F_{DE} > F_{AB} > F_{BC}$

Q65. MCQ (Single Correct Answer)

If an alpha particle with energy 7.7 MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is ____ m. (Atomic number of gold $= 79$ and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ in SI units)

Open solution

(A) 2.95×10^{-16} (B) 3.85×10^{-14} (C) 2.95×10^{-14} (D) 3.85×10^{-16}

Q66. MCQ (Single Correct Answer)

A current carrying solenoid is placed vertically and a particle of mass m with charge Q is released from rest. The particle moves along the axis of solenoid. If g is acceleration due to gravity then the acceleration (a) of the charged particle will satisfy :

Open solution

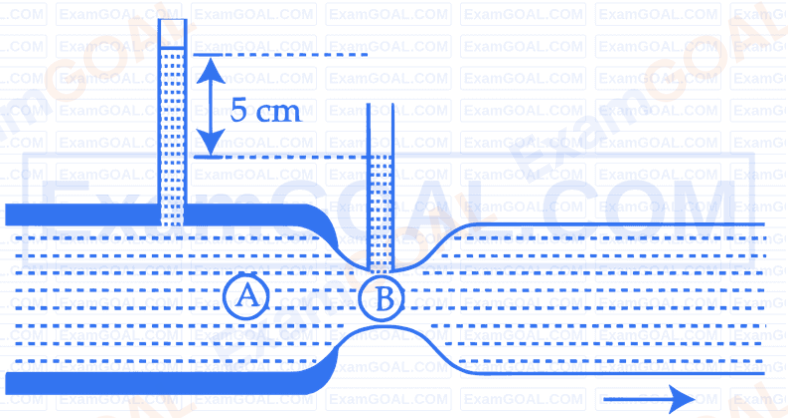
(A) $a > g$ (B) $a = 0$ (C) $0 < a < g$ (D) $a = g$

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Q67. MCQ (Single Correct Answer)

Open solution

Water flows through a horizontal tube as shown in the figure. The difference in height between the water columns in vertical tubes is 5 cm and the area of cross-sections at A and B are $6\ \mathrm{cm}^2$ and $3\ \mathrm{cm}^2$ respectively. The rate of flow will be ____ cm^3/s. (take $g = 10\ \mathrm{m/s}^2$)



(A) $100\sqrt{3}$ (B) $\frac{200}{\sqrt{3}}$ (C) $200\sqrt{6}$ (D) $200\sqrt{3}$

Q68. MCQ (Single Correct Answer)

Open solution

In a double slit experiment the distance between the slits is 0.1 cm and the screen is placed at 50 cm from the slits plane. When one slit is covered with a transparent sheet having thickness t and refractive index $n(=1.5)$, the central fringe shifts by 0.2 cm . The value of t is

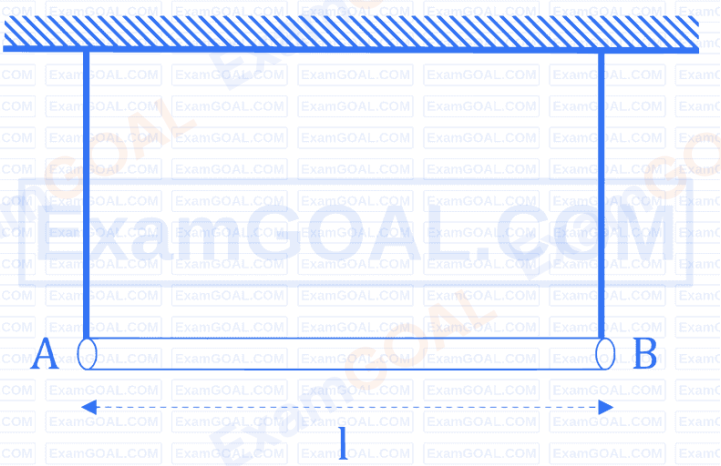
____ cm.

(A) 6.0×10^{-3} (B) 8×10^{-4} (C) 5.0×10^{-3} (D) 5.6×10^{-4}

Q69. MCQ (Single Correct Answer)

Open solution

A uniform rod of mass m and length l is suspended by means of two identical inextensible light strings as shown in figure. Tension in one string immediately after the other string is cut, is ____ . (g acceleration due to gravity)



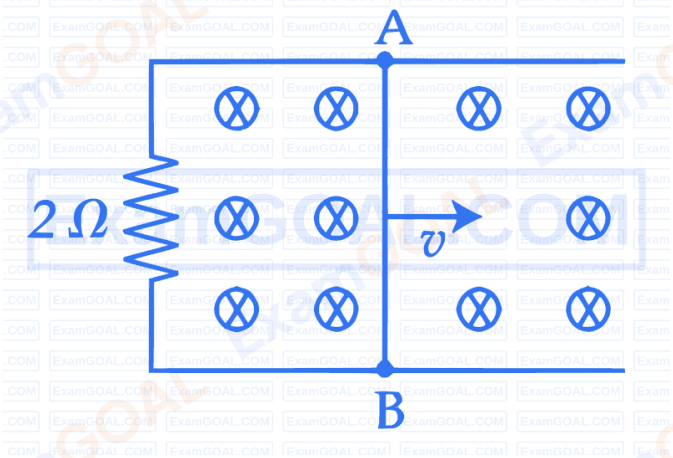
(A) $\mathrm{mg/s}$ (B) $mg/4$ (C) mg (D) $mg/2$

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Q70. MCQ (Single Correct Answer)

A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T . If the resistance of the total circuit is 2Ω then the force needed to move the rod towards right with constant speed (v) of $1.5\ \mathrm{m/s}$ is ____ N.

Open solution



(A) 5.7×10^{-2} (B) 7.5×10^{-3} (C) 5.7×10^{-3} (D) 7.5×10^{-2}

Q71. Numerical

10 mole of oxygen is heated at constant volume from $30^{\circ}\mathrm{C}$ to $40^{\circ}\mathrm{C}$. The change in the internal energy of the gas is ____ cal. (The molecular specific heat of oxygen at constant pressure, $C_P = 7\mathrm{cal/mol.}\,^{\circ}\mathrm{C}$ and $\mathrm{R} = 2\mathrm{cal.}/\mathrm{mol.}\,^{\circ}\mathrm{C.}$)

Open solution

Q72. Numerical

In a microscope the objective is having focal length $f_o = 2\ \mathrm{cm}$ and eye-piece is having focal length $f_e = 4\ \mathrm{cm}$. The tube length is 32 cm . The magnification produced by this microscope for normal adjustment is ____ .

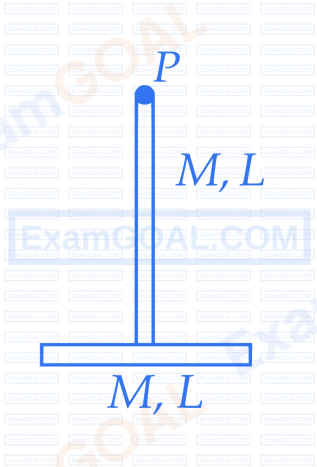
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Q73. Numerical

Two identical thin rods of mass M kg and length L m are connected as shown in figure. Moment of inertia of the combined rod system about an axis passing through point P and perpendicular to the plane of the rods is $\frac{x}{12}ML^2$ kg m^2. The value of x is ____ .

Open solution



Q74. Numerical

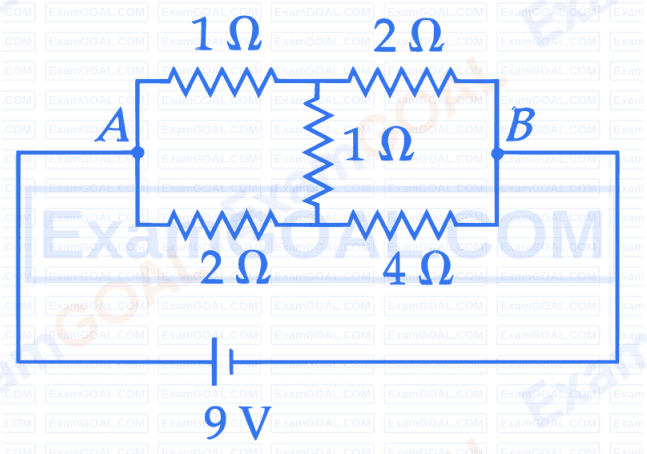
A collimated beam of light of diameter 2 mm is propagating along x-axis. The beam is required to be expanded in a collimated beam of diameter 14 mm using a system of two convex lenses. If first lens has focal length 40 mm , then the focal length of second lens is ____ mm.

Open solution

Q75. Numerical

The heat generated in 1 minute between points A and B in the given circuit, when a battery of 9 V with internal resistance of 1Ω is connected across these points is ____ J.

Open solution



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Answer Key

Chemistry

1. **C**	2. **D**	3. **B**	4. **D**	5. **B**
6. **B**	7. **C**	8. **C**	9. **A**	10. **B**
11. **C**	12. **A**	13. **A**	14. **B**	15. **B**
16. **A**	17. **B**	18. **D**	19. **A**	20. **A**
21. **8**	22. **70**	23. **13**	24. **6**	25. **20**

Mathematics

26. **A**	27. **B**	28. **A**	29. **D**	30. **C**
31. **B**	32. **B**	33. **B**	34. **A**	35. **B**
36. **C**	37. **C**	38. **C**	39. **D**	40. **B**
41. **A**	42. **C**	43. **A**	44. **D**	45. **D**
46. **1**	47. **1333**	48. **2**	49. **17**	50. **225**

Physics

51. **C**	52. **C**	53. **C**	54. **D**	55. **C**
56. **B**	57. **A**	58. **D**	59. **D**	60. **A**
61. **D**	62. **B**	63. **D**	64. **A**	65. **C**
66. **D**	67. **D**	68. **B**	69. **B**	70. **B**
71. **500**	72. **100**	73. **17**	74. **280**	75. **1080**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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